
JUnit Programming

Exercise 1: Setting Up JUnit

Scenario: You need to set up JUnit in your Java project to start writing unit tests.

Steps :

1. Create a new Java project in your IDE (e.g., IntelliJ IDEA, Eclipse).
2. Add the JUnit dependency to your project. If you are using Maven, add the following to your pom.xml:

```
<dependency>
  <groupId>junit</groupId>
  <artifactId>junit</artifactId>
  <version>4.13.2</version>
  <scope>test</scope>
</dependency>
```

3. Create a new test class in your project.

Solution Code :

```
import org.junit.Test;
import static org.junit.Assert.*;
```

```
public class MyFirstTest {

    @Test
    public void testSample() {
        assertEquals(4, 2 + 2);
    }
}
```

Output :

JUnit version 4.13.2

.

Time: 0.012

OK (1 test)

Exercise 3: Assertions in JUnit

Scenario: You need to use different assertions in JUnit to validate your test results.

Steps :

1. Write tests using various JUnit assertions.

Solution Code :

```
import org.junit.Test;
import static org.junit.Assert.*;
```

```
public class AssertionsTest {

    @Test
    public void testAssertions() {
        // Assert equals
        assertEquals(5, 2 + 3);

        // Assert true
        assertTrue(5 > 3);

        // Assert false
        assertFalse(5 < 3);

        // Assert null
        assertNull(null);

        // Assert not null
        assertNotNull(new Object());
    }
}
```

Output :

JUnit version 4.13.2

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Time: 0.008

OK (1 test)

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

Scenario: You need to organize your tests using the Arrange-Act-Assert (AAA) pattern and use setup and teardown methods.

Steps :

1. Write tests using the AAA pattern.
2. Use `@Before` and `@After` annotations for setup and teardown methods.

Solution Code :

```
import org.junit.Before;
import org.junit.After;
import org.junit.Test;
import static org.junit.Assert.*;

public class CalculatorTest {

    private Calculator calculator;

    @Before
```

```
public void setUp() {
    // Arrange: runs before every test
    calculator = new Calculator();
    System.out.println("Setting up Calculator instance");
}

@Test
public void testAddition() {
    // Arrange
    int a = 10;
    int b = 5;

    // Act
    int result = calculator.add(a, b);

    // Assert
    assertEquals(15, result);
}

@Test
public void testSubtraction() {
    // Arrange
    int a = 10;
    int b = 5;

    // Act
    int result = calculator.subtract(a, b);

    // Assert
    assertEquals(5, result);
}

@After
public void tearDown() {
    // Cleanup after every test
    calculator = null;
    System.out.println("Tearing down Calculator instance");
}
}
```

Output :

```
Setting up Calculator instance
Tearing down Calculator instance
Setting up Calculator instance
Tearing down Calculator instance
```

JUnit version 4.13.2

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Time: 0.010

OK (2 tests)
